

1.8 Quiz Corrections: Measurement

For skills you earned lower than 9 or 10, complete the review problems below.

Reference (provided). $\text{average speed} = \frac{\text{distance}}{\text{time}}$

1 m $\approx$ 3.28 ft 1 mi = 5280 ft 1 km = 1000 m 1 m = 100 cm 1 cm = 10 mm

1 h = 3600 s 1 min = 60 s 1 gal $\approx$ 3.785 L

1. Write down the number of significant digits in each measurement.

(a) 0.0480 m

(c) 5.0 s

(e) 0.000 03 kg

(b) 409 g

(d) 20.40 cm

(f) 1.67×10^{-27} kg

2. Round each value to three significant figures. Copy the calculator display followed by three dots, then round: $\sqrt{2} = 1.4142135 \dots \approx 1.41$

(a) $\sqrt{7}$

(b) The height of Mont Blanc, measured in 2023, 4805.59 m

3. Write in scientific notation, rounded to three significant figures.

(a) The average distance from the Earth to the Sun, about 149,597,871 km

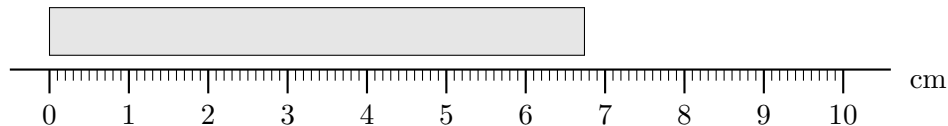
(b) The radius of a hydrogen atom, 0.000 000 000 052 917 7 m

4. Write each value as an ordinary decimal number.

(a) The thickness of a sheet of paper, 1.0×10^{-4} m

(b) The average distance from the Earth to the Moon, 3.84×10^5 km

5. The bar below is measured with a centimeter ruler marked in millimeters.



- (a) Record the length of the bar. Include one estimated digit — the last digit, the one you are not certain of.
- (b) How many significant figures does your measurement have?
- (c) Write your measurement in millimeters.
6. Convert between units, rounding to three significant figures. Write each conversion as a single chain of fractions and show the units cancelling.
- (a) The Burj Khalifa is 828 meters tall. What is its height in feet?
- (b) Italy's Frecciarossa 1000 high-speed train runs at 300 kilometers per hour. What is this speed in meters per second?
7. A skydiver falling belly-down reaches a top speed of about 55 meters per second. What is this speed in miles per hour?